

ModernDataSciEng Platform

A scalable, governed data platform — trusted, single-source-of-truth datasets across the business

Reference architecture · 14 pages · Knowledge Loop

Document	ModernDataSciEng Platform — Project Report	
Version	v2.4 (FY26)	
Synthetic data	All numbers, schemas, pipelines and dashboards are hypothetical	
Compliance	GDPR Reg. 2016/679 — PII tagged + masked via Unity Catalogue	
Public repo	github.com/testdemoqwenai2025-creator/DemoAppDataSci	

Private repo	github.com/testdemoqwenai2025-creator/AppDataSci-Advanced
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1. Executive Summary

The ModernDataSciEng Platform is a synthetic reference implementation of a modern, governed data platform for a hypothetical omnichannel retailer (ModernDataSciEng Ltd). Every number, schema, pipeline and dashboard in this document is hypothetical and illustrative — the platform exists to show how a small, capable data engineering team partners with stakeholders, analysts and platform teams to onboard new sources, improve quality and ship governed analytics at scale. The platform spans Snowflake, Databricks, dbt, Tableau, Fivetran, Hightouch, Airflow and Unity Catalogue — every layer is purpose-built, observable and CI/CD-driven.

This document is the canonical project report. It summarises the architecture, technology stack, the fourteen pages of the application, the Knowledge Loop that distinguishes this platform from static reference architectures, the GDPR compliance posture, and the author's thoughts on ever-evolving technologies that should be considered for the next 24-36 months. It is the entry point for new engineers, auditors and interested external parties who want to understand the platform in one sitting.

Headline outcomes (synthetic)

Metric	Value	Trend
Pipeline freshness (P95)	9 min	−37% vs FY24
Data trust score	98.4 / 100	+4.2 pts
Active governed datasets	268	+54 YoY
Self-service analysts	312	+118 YoY
Annual platform cost	£2.8M	−19% / TB
Data incidents (P0/P1)	3	−71% vs FY24

2. Reference Architecture

The platform is built as four logical planes — sources, lakehouse, serving warehouse and consumption — wired together with orchestration, governance and CI/CD. The canonical data flow is unidirectional: source systems are ingested via Fivetran into the Bronze layer (append-only Delta on ADLS Gen2 / S3); PySpark + Delta MERGE conform the data into Silver (deduplicated, validated, joined to SCD2 dimensions); dbt + PySpark DLT build the Gold dimensional marts; Snowflake serves them as SECURE VIEWS with row-level security; Tableau + Hightouch consume and activate the data.

Each layer has a single, hard-enforced responsibility. Bronze is append-only and schema-on-read; Silver is conformed via idempotent MERGE on business keys; Gold is dimensional and BI-facing. This separation makes the platform debuggable (every problem has a layer), rerunnable (no layer corrupts upstream), and cost-optimised (each layer can be tuned independently — Z-ORDER on Silver, cluster keys on Gold, lifecycle rules on Bronze).

Layer-by-layer responsibilities

Layer	Technology	Role
Sources	Shopify, Salesforce, Stripe, NetSuite, Klaviyo, Snowplow, Asana, Mailchimp, Google Ads, SAP, Amazon, WooCommerce, HubSpot, Zendesk, etc.	System of record, generate events
Ingestion	Fivetran (ELT) + Snowplow (event pipeline) + Kafka (streaming)	Schema-on-read CDC into Bronze, no transformations here
Lakehouse	Azure ADLS Gen2 / S3 + Delta Lake	Single open storage format, ACID + time travel + Z-ORDER
Compute	Databricks (Spark, PySpark, Photon, SQL warehouses)	Bronze → Silver → Gold transforms + ML feature engineering

Warehouse	Snowflake (multi-cluster warehouses)	BI-grade SQL serving, RLS, secure sharing, reverse-ETL API
Transformation	dbt Core + dbt Cloud + MetricFlow semantic layer	Staging → Intermediate → Marts → Serving, tests & docs
Orchestration	Airflow (batch) + Dagster (asset-based)	Idempotent DAGs, SLAs, retries, circuit-breakers
Analytics	Tableau Server + Tableau Catalog + embedded APIs	Governed dashboards, RLS, certified datasets, alerting
Reverse-ETL	Hightouch	Push governed audiences back into CRM / CDP / ad platforms
Governance	Unity Catalogue + Monte Carlo + OpenLineage + Immuta	PII tagging, lineage, DQ, RBAC, audit & masking
IaC / CI/CD	Terraform + GitHub Actions + dbt slim CI	Reproducible environments, state-aware model promotion

3. The Fourteen Pages of the Application

The application is a multi-page application (MPA). Every page is dedicated to a specific technology or scope, addressable by URL hash ([#/snowflake](#), [#/databricks](#), etc.). The first ten pages are the architecture pages — sources, lakehouse, warehouse, transformation, orchestration, analytics, activation, governance, CI/CD. The last four pages form the Knowledge Loop, a unique feature that turns the platform from a static reference into a living, citation-driven organism.

#	Page	Scope
1	Platform Overview	Executive KPIs, mini architecture diagram, design principles
2	Reference Architecture	End-to-end diagram, layer-by-layer responsibilities, environment matrix
3	Fivetran & Hightouch	ELT ingestion from 14 sources, reverse-ETL activation into 5 audiences
4	Databricks Lakehouse	PySpark, Delta Lake, Medallion Bronze/Silver/Gold, Photon runtime
5	Snowflake & SQL	Multi-cluster warehouses, RBAC, Gold SECURE VIEW, cluster keys
6	dbt & Dimensional Modelling	312 models, 1,184 tests, SCD2 snapshots, MetricFlow semantic layer, slim CI
7	Airflow & Dagster	Idempotent DAGs, SLA monitoring, partition-aware backfill
8	Tableau & Analytics	Certified datasets, RLS via session context, sample dashboards
9	Data Governance & Observability	Unity Catalogue grants, PII tags, Monte Carlo monitors, OpenLineage
10	Git, CI/CD & DevOps	Trunk-based flow, GitHub Actions, Terraform, OIDC, FinOps chart
11	About & Compliance	Mission, synthetic-data disclaimer, GDPR rights table, contact, repo links
12	Knowledge Hub	12 ADRs, 8 patterns, 4 trade-off matrices — the why behind every choice
13	Live Dashboard	Synthetic real-time observatory — pipeline runs, cost, anomalies, what-if simulator
14	Evolution Timeline	Versioned history v1.0 → v2.4 → v3.0, tech radar, 4-horizon future roadmap
15	Research Papers	16 academic foundations, citation graph, paper-to-platform mapping

4. The Knowledge Loop — what makes this platform different

Most reference architectures are read-once diagrams. You read them, you leave. The Knowledge Loop turns the ModernDataSciEng Platform into a living organism by closing a circuit between four pages: Research → Knowledge → Architecture → Dashboard → back to Research. Papers inspire patterns; patterns become architecture; architecture is observed in the live dashboard; observations drive new research (new papers to read, new ADRs to write, new patterns to try). The platform is never finished — it's an ongoing conversation between theory and production.

The loop in detail

Step	Page	What it does
1	Research Papers	16 academic foundations (MapReduce, RDDs, Delta Lake, Kimball, Lakehouse, OpenLineage). Interactive citation
2	Knowledge Hub	12 Architecture Decision Records (ADRs) — context, decision, consequences, alternatives considered. 8 reusable
3	Architecture pages	Ten pages (Snowflake, Databricks, dbt, Tableau, Fivetran/Hightouch, Airflow, Unity Catalogue, CI/CD) — the ac
4	Live Dashboard	Synthetic real-time observatory — pipeline runs tick in every 3s, credits burn, anomalies flow. What-if simulator p

The honest framing: the loop is never closed perfectly. Some academic foundations don't have ADRs yet; some patterns aren't documented; some dashboard observations don't make it back to research. The loop is an aspiration — a discipline the team practices — not a state it achieves.

5. Governance & GDPR Compliance

The platform treats governance as a first-class engineering concern, not a compliance afterthought. Unity Catalogue provides column-level RBAC and PII tagging; Monte Carlo detects freshness, volume and schema-drift anomalies; OpenLineage ties everything together end-to-end. PII columns are tagged, surfaced through `_masked` views with role-based redaction, and access is audited.

GDPR rights operationalised as engineering controls

GDPR Right	Engineering Implementation
Right of access (Art. 15)	Unity Catalogue audit log + access reviews
Right to rectification (Art. 16)	dbt snapshot history + MERGE on business key
Right to erasure (Art. 17)	Delta DELETE + VACUUM + downstream dbt re-run
Right to data portability (Art. 20)	Snowflake secure sharing + Parquet export
Right to object (Art. 21)	Opt-out registry synced from Zendesk → Hightouch
Right to restrict processing (Art. 18)	RLS policy + Immuta masking toggle

Data subject requests (DSARs) are acknowledged within 72 hours and fulfilled within 30 days, in line with GDPR Article 12. All PII columns are tagged in Unity Catalogue; the same tags drive Hightouch masking and Tableau access. The synthetic data disclaimer applies throughout: no real personal data is processed, stored or transmitted anywhere in this reference implementation.

6. Thoughts on Ever-Evolving Technologies

What follows is opinionated, dated, and deliberately provocative. It is a watch list for the next 24-36 months, not a commitment list. The full document lives in `FUTURE_TECH.md` in the repository; this section summarises the highest-confidence bets.

Storage & formats

Delta Uniform (Iceberg interop) will be table stakes by FY27 — plan for it now. DuckDB-in-CI is a Q2 priority to validate dbt models against sample data without provisioning a warehouse. Polars replaces Pandas in our PySpark ad-hoc notebooks by FY26 H2. Confidential compute on object stores (SGX, SEV-SNP, TDX) is niche today but will be mainstream by FY28; build the platform so it can be enabled per-table when needed.

Compute

Snowflake Cortex and Databricks AI Functions are putting LLM inference inside the warehouse, eliminating the ETL-into-ML-system pattern for many use cases. By FY27, most NLP-on-customer-feedback and PII-detection workloads will run inside the warehouse without a separate ML pipeline. Quantum-safe encryption is low priority for FY26 but watch vendor PQC roadmaps; plan for a migration window starting FY28.

Modelling & orchestration

Streaming-native transformations (Spark readStream + Delta CDF + Kafka) blur the batch/streaming line; streaming-first Bronze moves from trial to adopt in FY26 Q2. Asset-based orchestration (Dagster) is becoming the de-facto mental model — Airflow 3.0 adopts it. The plan is to migrate 50% of DAGs to Dagster assets by FY26 H2, 100% by FY27. OpenTelemetry everywhere will make observability vendor-neutral by FY27.

AI & agents

The biggest bet is the agentic operating model — DQ triage agents, schema-drift PR agents, cost optimisation agents, semantic Q&A; agents. The hypothesis: agents reduce on-call load by 60% and pipeline delivery from 2 weeks to 2 days. The risk: agents making schema changes autonomously. The mitigation: human approval for every PR, immutable audit log, daily LLM-token cost cap, PII never leaves the warehouse. The full design lives in AGENTIC_WORKFLOW.md in the repository.

Libraries & languages

Rust is becoming the implementation language for high-performance data tooling (Polars, DataFusion, Apache Arrow core, Vector.dev). The modern Python stack is consolidating around Polars + Pydantic v2 + uv + ruff + maturin. The frontend stack (Next.js + TypeScript + Tailwind 4 + shadcn/ui + Bun) is settled — watch Million.js for React performance and TanStack Start for full-stack TS framework disruption.

7. Closing — how to use this platform

This report is the entry point. The application is the platform. Read this PDF for the executive summary; browse the public repository for the source code; clone the private repository to make modifications. The sync workflow mirrors every push to the private repo into the public one — anyone can preview the latest state of the platform without signing an NDA. If you find an architectural choice you'd challenge, write an ADR. If you find a research paper that should be in the citation graph, open a PR. The platform evolves because the team is honest about which assumptions held and which didn't.

Synthetic data disclaimer: every number, schema, pipeline, dashboard and business name in this report is synthetic and hypothetical. Any resemblance to real companies, persons, products or events is coincidental. If you would like to validate any figure shown here against a real business, please contact the team — the synthetic numbers are illustrative only.